

IIT Roorkee × Futurense
PG Certificate Program in AI/GenAI Powered
Cybersecurity
Capstone Project — Cohort 1

AlertMind — AI-Assisted Mini Security Operations Centre

Submitted by
Soumil Verma
Karnamughilhan R
Group 10 — Defense Matrix

CHAPTER 1

1. Introduction

1.1 Background

The rapid growth of digital transformation, cloud computing, and interconnected enterprise environments has significantly expanded the cyber threat landscape. Organizations across industries are increasingly targeted by sophisticated cyberattacks such as ransomware, phishing campaigns, credential theft, insider threats, and advanced persistent threats (APTs). To defend against these attacks, Security Operations Centers (SOCs) continuously monitor enterprise systems, analyze security events, investigate suspicious activities, and coordinate incident response.

Modern Security Information and Event Management (SIEM) platforms, including Wazuh, Microsoft Sentinel, Splunk Enterprise Security, IBM QRadar, and Elastic Security, collect and correlate security logs generated by operating systems, applications, network devices, and cloud environments. These platforms enable security analysts to detect malicious behavior by applying detection rules, threat intelligence, and behavioral analytics.

Despite these capabilities, SOC analysts face a growing challenge commonly referred to as **alert fatigue**. Large enterprise environments may generate thousands of security alerts every day, many of which are low-priority events or false positives. Analysts must manually review alerts, identify relevant MITRE ATT&CK techniques, determine appropriate investigation steps, and prepare incident response recommendations. This manual workflow increases investigation time, introduces inconsistencies between analysts, and reduces the efficiency of security operations.

Recent advances in Large Language Models (LLMs) have created new opportunities to improve SOC workflows. Rather than replacing human analysts, LLMs can function as intelligent assistants capable of interpreting alerts, summarizing security events, mapping adversary behavior to the MITRE ATT&CK framework, recommending investigation steps, and drafting communications for affected stakeholders. When combined with appropriate guardrails and human oversight, AI-assisted SOC solutions have the potential to reduce analyst workload while maintaining responsible decision-making.

1.2 Motivation

The motivation behind this project originated from the practical challenges encountered during SOC alert analysis. Traditional SIEM deployments effectively detect suspicious activities but often require analysts to spend considerable time understanding each alert before beginning an investigation.

For entry-level SOC analysts, this process can be both time-consuming and inconsistent. Different analysts may interpret identical alerts differently depending on their experience and familiarity with attack techniques. Furthermore, increasing alert volumes reduce the time available for detailed investigation of genuinely critical incidents.

The emergence of Generative AI presents an opportunity to augment SOC operations by assisting analysts with repetitive analytical tasks while preserving human decision-making authority.

1.3 Project Overview

This capstone project presents **AlertMind**, an AI-assisted Mini Security Operations Centre (SOC) that integrates the Wazuh SIEM platform with a Large Language Model (LLM) to enhance incident triage and analysis.

The project consists of three primary components:

1. **Security Monitoring Layer:** A **Wazuh-based SIEM** deployed on Ubuntu Server collects and analyzes security events generated by Windows Server 2019, Linux audit logs, and AWS CloudTrail. Sysmon was configured on the Windows endpoint to provide detailed process creation and system activity logs.
2. **Detection Engineering Layer:** A set of **custom detection rules** was developed to identify suspicious activities including Windows command execution, PowerShell misuse, registry persistence, scheduled task creation, account creation, Windows service installation, failed logins, and other security-relevant behaviors. The detections were mapped to corresponding **MITRE ATT&CK** techniques, and two operational dashboards were created to visualize security events and attack trends.
3. **AI-Assisted Analysis Layer:** A **LangChain**-based application utilizing the **Groq**-hosted Llama 3.1 8B Instant model was developed to process exported Wazuh alerts. The assistant automatically generates concise alert summaries, maps alerts to MITRE ATT&CK techniques, recommends investigation queries, suggests analyst actions, and drafts security notifications. Multiple guardrails, including prompt sanitization, response

logging, human review requirements, and non-automated operation, were implemented to ensure responsible AI usage.

The effectiveness of the proposed solution was evaluated by executing Atomic Red Team attack simulations before and after integrating the AI assistant. Performance was assessed using Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and qualitative improvements in analyst workflow efficiency.

1.4 Project Contributions

The primary contributions of this capstone project are summarized below:

- Design and deployment of a functional Security Information and Event Management (SIEM) environment using Wazuh.
- Development and validation of 12 custom detection rules aligned with the MITRE ATT&CK framework.
- Implementation of two operational dashboards for SOC monitoring and attack visualization.
- Creation of three Incident Response (IR) playbooks based on the NIST SP 800-61 Rev. 2 incident response lifecycle.
- Design and implementation of the AlertMind AI Assistant using LangChain and the Groq-hosted Llama 3.1 8B Instant model.
- Implementation of responsible AI guardrails including prompt sanitization, audit logging, and mandatory human review.
- Experimental evaluation of AI-assisted alert triage using Atomic Red Team attack simulations.
- Comparative analysis of baseline and AI-assisted SOC workflows using MTTD and MTTR measurements

CHAPTER 2

Problem Statement, Objectives, Scope and Deliverables

2.1 Problem Statement

In a typical SOC environment, analysts receive alerts containing raw event data, rule descriptions, process information, user details, timestamps, and other security artifacts. Before an analyst can determine whether an event represents malicious activity, they must manually perform several tasks, including:

- Understanding the alert context.
- Identifying the relevant MITRE ATT&CK technique.
- Determining whether the observed behavior is expected or suspicious.
- Planning investigation queries.
- Recommending containment or response actions.
- Preparing notifications or incident documentation.

As the number of monitored endpoints increases, the volume of generated alerts also grows. This often results in **alert fatigue**, where analysts spend a significant amount of time reviewing repetitive or low-priority alerts instead of focusing on genuinely critical incidents.

Although recent advances in Large Language Models (LLMs) demonstrate considerable potential for assisting cybersecurity analysts, unrestricted use of AI introduces additional concerns, including inaccurate recommendations, hallucinated information, privacy risks, and excessive reliance on automated decision-making. This project addresses these challenges by developing an AI-assisted Security Operations Centre prototype named **AlertMind**, which integrates a traditional SIEM platform with an LLM-based analyst assistant.

2.2 Project Objectives

The primary objective of this project is to design, implement, and evaluate an AI-assisted Security Operations Centre capable of improving analyst efficiency without compromising responsible security operations.

The specific objectives of the project are listed below:

1. Deploy a functional Security Information and Event Management (SIEM) platform using Wazuh.
2. Configure Windows and Linux endpoints for centralized security log collection using Sysmon, Windows Event Logs, and Linux audit logs.
3. Develop and validate at least ten custom detection rules capable of identifying common attacker techniques.
4. Map custom detection rules to the MITRE ATT&CK framework to improve attack classification and investigation.
5. Design operational dashboards for real-time SOC monitoring and attack visualization.
6. Develop Incident Response playbooks aligned with the NIST SP 800-61 Rev. 2 incident response lifecycle.
7. Design and implement an AI-assisted Tier-1 SOC analyst using LangChain and a Large Language Model.
8. Implement responsible AI guardrails including prompt sanitization, audit logging, response logging, and mandatory human approval before any operational decision.
9. Evaluate the effectiveness of AI-assisted alert triage by executing Atomic Red Team attack simulations and comparing baseline and AI-assisted workflows using MTTD and MTTR.
10. Produce comprehensive project documentation suitable for deployment demonstration and academic evaluation.

2.3 Project Scope

The project was implemented within a VMware-based laboratory environment consisting of Windows Server 2019, Ubuntu Server running Wazuh, a dedicated Ubuntu virtual machine hosting the AlertMind AI Assistant, Kali Linux, and AWS CloudTrail.

The scope includes SIEM deployment, endpoint monitoring, custom detection engineering, dashboard development, incident response documentation, AI-assisted alert analysis, and performance evaluation. Automated incident response actions and enterprise-scale deployment were intentionally excluded, as the project focuses on demonstrating AI-assisted analyst support within a controlled environment.

2.4 Project Deliverables

The final deliverables achieved during the capstone project are summarized here:

Deliverable	Status
SOC Architecture Document	✓
Working Wazuh SIEM with Live Log Ingestion	✓
12 Custom Detection Rules and Sigma Rule Pack	✓
Two Operational Dashboards	✓
Three Incident Response Playbooks	✓
AlertMind AI Assistant	✓
Prompt Library and AI Guardrails	✓
Atomic Red Team Evaluation	✓

2.5 Success Criteria

The project is considered successful if it demonstrates:

- Successful deployment of a functional SIEM with centralized log collection.
- Accurate detection of simulated attacks using custom detection rules.
- Effective visualization through operational dashboards.
- Practical Incident Response documentation.
- AI-assisted alert analysis with responsible guardrails.
- Improved analyst workflow demonstrated through the evaluation of MTTD and MTTR.

CHAPTER 3

Literature Review

3.1 Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) systems play a vital role in modern cybersecurity by collecting, normalizing, correlating, and analyzing security events generated across an organization's infrastructure. They provide centralized visibility into activities occurring on endpoints, servers, network devices, and cloud environments, enabling security analysts to detect potential threats and investigate incidents more efficiently. SIEM platforms also support compliance reporting, log retention, and real-time monitoring, making them an essential component of Security Operations Centres (SOCs).

3.2 Wazuh as an Open-Source SIEM Platform

Wazuh is an open-source security platform that combines host-based intrusion detection, log management, file integrity monitoring, vulnerability detection, and security event correlation within a unified architecture. It consists of a central manager, endpoint agents, an indexer, and a web-based dashboard for monitoring and visualization.

One of Wazuh's major strengths is its ability to collect security logs from multiple operating systems while supporting custom detection rules based on XML decoders and Sigma-inspired detection methodologies. This flexibility enables organizations to tailor detection logic according to their security requirements.

3.3 MITRE ATT&CK Framework

The MITRE ATT&CK framework is a globally recognized knowledge base that categorizes adversarial tactics, techniques, and procedures (TTPs) observed in real-world cyberattacks. Instead of focusing solely on malware families or threat actors, ATT&CK describes how attackers achieve their objectives across different stages of an attack lifecycle.

Security vendors and SOC teams widely use ATT&CK to map detection rules, improve threat hunting, identify defensive gaps, and standardize incident reporting. Mapping detections to ATT&CK techniques also improves communication among analysts by providing a common language for describing attacker behavior.

3.4 Detection Engineering Using Sigma Rules

Detection engineering is the process of developing and maintaining detection logic capable of identifying malicious behavior within security logs. Effective detection rules must balance sensitivity and specificity to minimize both false negatives and false positives.

Sigma is an open and vendor-independent rule format that enables analysts to describe security detections using a standardized syntax. Although Sigma rules require platform-specific conversion before deployment, they promote reusable detection logic across multiple SIEM solutions. The detection strategy implemented in AlertMind follows Sigma-inspired principles by focusing on attacker behaviors rather than specific malware signatures

3.5 Atomic Red Team for Detection Validation

Atomic Red Team is an open-source adversary emulation framework that provides small, repeatable tests corresponding to MITRE ATT&CK techniques. These tests enable defenders to validate security controls without deploying full attack scenarios or complex penetration testing environments.

In this project, selected Atomic Red Team techniques were executed within the laboratory environment to validate custom detection rules and measure the effectiveness of the proposed AI-assisted workflow. The use of Atomic Red Team ensured that the implemented detections were evaluated against standardized attack simulations rather than theoretical assumptions.

3.6 Large Language Models in Security Operations

Recent advances in Large Language Models (LLMs) have introduced new opportunities for improving SOC workflows. Models such as GPT, Llama, and Mistral demonstrate strong capabilities in natural language understanding, reasoning, and structured text generation, making them suitable for assisting security analysts during alert investigation.

Rather than replacing traditional detection mechanisms, LLMs complement existing SIEM platforms by interpreting alerts, summarizing technical information, identifying relevant ATT&CK techniques, recommending investigation steps, and generating incident documentation. These capabilities can reduce repetitive manual work while enabling analysts to focus on higher-priority investigations.

CHAPTER 4

System Architecture

4.1 Overview

The implementation was developed in a VMware-based virtual laboratory consisting of Windows Server 2019, Ubuntu Server hosting the Wazuh platform, a dedicated Ubuntu virtual machine running the AlertMind AI Assistant, Kali Linux for security testing, and AWS CloudTrail for cloud activity monitoring. Together, these components simulate a small enterprise.

4.2 System Components

The proposed architecture consists of five major components that work together to provide end-to-end security monitoring and AI-assisted analysis.

4.2.1 Endpoint Layer

The endpoint layer includes Windows Server 2019 and Ubuntu Linux systems monitored by Wazuh agents.

The Windows endpoint was configured with **Sysmon**, enabling detailed logging of process creation, command-line execution, service installation, scheduled tasks, registry modifications, and user account activities. The Ubuntu endpoint utilized the Linux auditing subsystem to generate security logs related to authentication events and system activities. These endpoints continuously forwarded security events to the Wazuh Manager for centralized analysis.

4.2.2 SIEM Layer

The Security Information and Event Management (SIEM) layer is powered by **Wazuh**, which serves as the central monitoring platform. The Wazuh Manager receives security logs from registered agents, applies custom detection rules, correlates events, and stores processed alerts within the OpenSearch Indexer. The Wazuh Dashboard provides a web-based interface for monitoring alerts, visualizing attack trends, and reviewing security events in real time.

4.2.3 Threat Simulation Layer

To validate the effectiveness of the implemented detection rules, **Atomic Red Team** was used to emulate real-world adversary techniques. Atomic tests corresponding to selected MITRE ATT&CK techniques were executed on the Windows endpoint to generate controlled attack

scenarios. The resulting events were captured by Sysmon, forwarded to Wazuh, and analyzed by the SIEM.

4.2.4 AI Analysis Layer

The AlertMind AI Assistant forms the intelligence layer of the proposed architecture. It was developed using the **LangChain** framework and integrates with the **Groq-hosted Llama 3.1 8B Instant** model.

Instead of processing raw endpoint logs directly, the assistant analyzes structured JSON alerts exported from Wazuh. For each alert, the assistant generates:

- A concise alert summary.
- MITRE ATT&CK technique mapping.
- Suggested investigation queries.
- Recommended analyst actions.
- A draft security notification.

This workflow reduces the time required for initial alert triage while ensuring that the final decision remains with the security analyst.

4.2.5 Security and Governance Layer

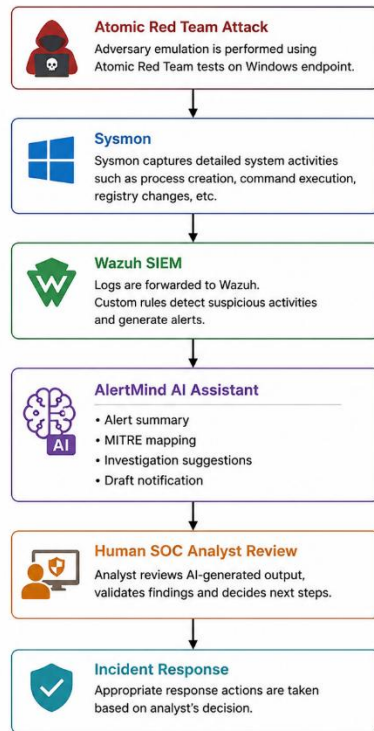
To ensure responsible use of Generative AI, several security controls were incorporated into the AlertMind assistant. User prompts are sanitized before processing, while all prompts and generated responses are recorded in dedicated log files to provide an audit trail. The assistant operates strictly as a decision-support tool and is not authorized to perform automated containment or remediation actions. Every recommendation must be reviewed and approved by a human analyst before implementation.

4.3 System Workflow

- The operational workflow of AlertMind consists of the following steps:
- Security events are generated on monitored Windows and Linux endpoints.
- Wazuh agents collect endpoint logs and securely forward them to the Wazuh Manager.
- The Wazuh Manager processes incoming events using custom detection rules and correlates them with predefined security policies.
- Detected alerts are indexed and displayed through the Wazuh Dashboard.
- Structured alert data is provided to the AlertMind AI Assistant.

- The assistant analyzes the alert and generates investigation guidance, MITRE ATT&CK mappings, recommended actions, and a draft notification.
- A security analyst reviews the AI-generated recommendations before making any operational decision

4.4 Overall Architecture



Component	Purpose
Windows Server 2019	Generates Windows security events and Sysmon logs
Ubuntu Server	Hosts the Wazuh Manager, Indexer, and Dashboard
Ubuntu AI VM	Runs the AlertMind AI Assistant
Kali Linux	Executes security testing and validation
AWS CloudTrail	Provides cloud activity logs
Wazuh	Collects, correlates, and analyzes security events
Sysmon	Generates detailed Windows telemetry

Component	Purpose
Atomic Red Team	Simulates adversary techniques
LangChain	Orchestrates the LLM workflow
Groq (Llama 3.1 8B Instant)	Performs AI-assisted alert analysis

CHAPTER 5

Experimental Environment and System Setup

5.1 Overview

A virtualized laboratory environment was developed to implement and evaluate the AlertMind AI-Assisted Security Operations Centre (SOC). The environment was created using **VMware Workstation** to simulate a small enterprise network consisting of Windows, Linux, and cloud log sources. This setup enabled centralized log collection, attack simulation, detection validation, and AI-assisted alert analysis without affecting production systems.

The laboratory consisted of four virtual machines and one cloud log source, allowing the complete security monitoring workflow to be tested in a controlled environment.

5.2 Hardware and Software Configuration

The project was implemented on a Windows 11 host machine with the specifications:

Component	Specification
Host Operating System	Windows 11
Processor	AMD Ryzen 7 Series
Memory	16 GB DDR4 RAM
Virtualization Platform	VMware Workstation
Available Storage	249 GB Free Disk Space

5.3 Virtual Laboratory Environment

The virtual environment consisted of four systems, each assigned a specific role within the SOC architecture.

Virtual Machine	Operating System	Purpose
Ubuntu Server	Ubuntu Server 24.04 LTS	Wazuh Manager, Indexer and Dashboard
Windows Server	Windows Server 2019	Endpoint monitored using Sysmon and Wazuh Agent
Ubuntu AI	Ubuntu Desktop	AlertMind AI Assistant

5.4 Software Components

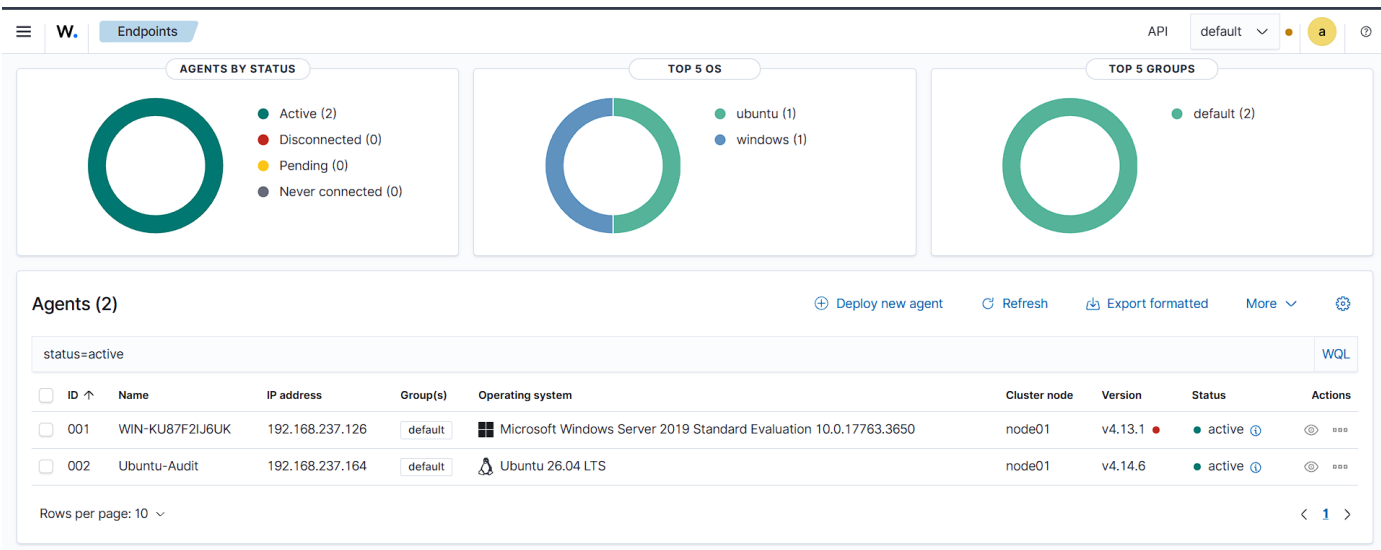
Several open-source tools and frameworks were integrated to build the AlertMind platform.

Software	Purpose
Wazuh	Security Information and Event Management (SIEM)
OpenSearch	Alert indexing and storage
Sysmon	Windows system activity monitoring
Wazuh Agent	Endpoint log collection
Atomic Red Team	Attack simulation
LangChain	LLM application framework

Software	Purpose
Groq API (Llama 3.1 8B Instant)	AI-assisted alert analysis
Python	Application development
AWS CloudTrail	Cloud activity logging

5.5 Laboratory Configuration

- The Wazuh Manager was deployed on the Ubuntu Server virtual machine, where it collected and processed security events received from the monitored endpoints. The Windows Server 2019 virtual machine was configured with the Wazuh Agent and Microsoft Sysmon to generate detailed telemetry, including process creation events, command execution, registry modifications, and authentication logs.
- A separate Ubuntu virtual machine hosted the AlertMind AI Assistant to ensure that AI processing remained isolated from the SIEM infrastructure. This separation improved modularity and prevented the AI application from affecting the performance of the Wazuh Manager.
- Additionally, AWS CloudTrail logs were integrated into the monitoring environment to demonstrate cloud activity monitoring. Custom detection rules were validated by executing Atomic Red Team attack simulations directly on the Windows Server endpoint.



5.6 Development Environment

The AlertMind assistant was developed using Python and the LangChain framework. The application processes structured JSON alerts exported from Wazuh and communicates with the Groq-hosted **Llama 3.1 8B Instant** model to generate AI-assisted analysis.

The assistant provides:

- Alert summaries
- MITRE ATT&CK mapping
- Investigation queries
- Recommended analyst actions
- Draft security notifications

To support responsible AI usage, prompt sanitization, prompt logging, response logging, and mandatory human review were incorporated into the application design.

CHAPTER 6

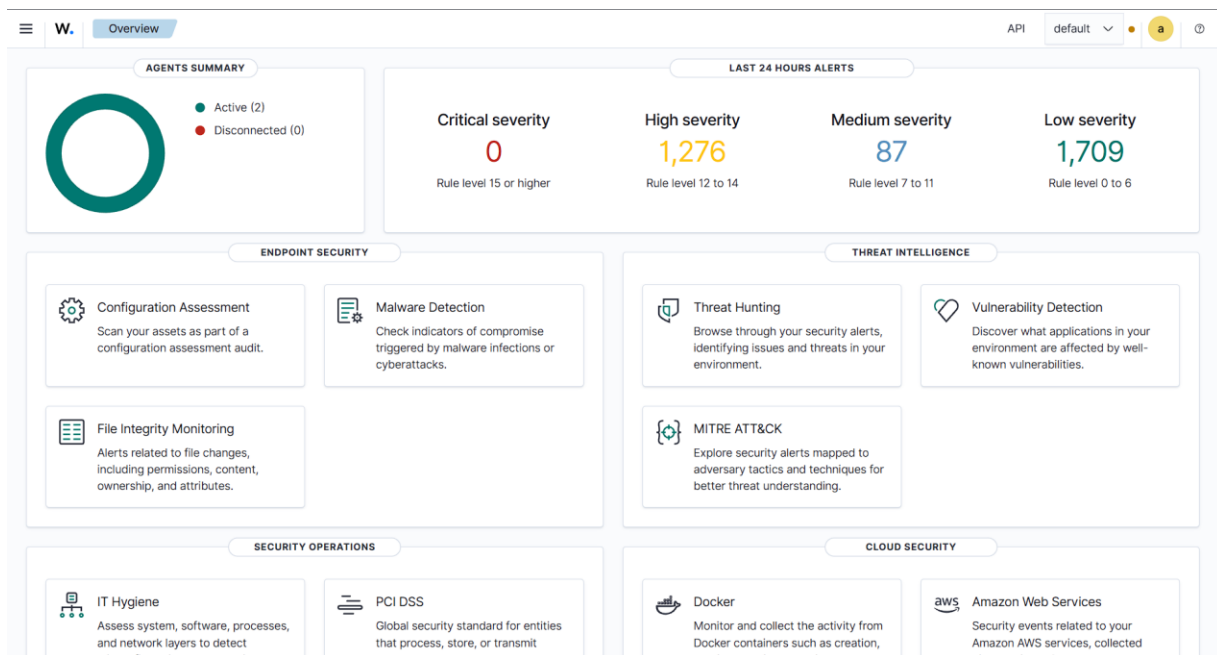
SIEM Deployment and Log Collection

6.1 Overview

The Security Information and Event Management (SIEM) platform forms the core of the AlertMind Security Operations Centre. Wazuh was selected as the primary SIEM solution due to its open-source architecture, centralized log management capabilities, support for custom detection rules, and integration with the MITRE ATT&CK framework. The deployment enables continuous monitoring of endpoint activities and provides the foundation for AI-assisted alert analysis.

6.2 Wazuh SIEM Deployment

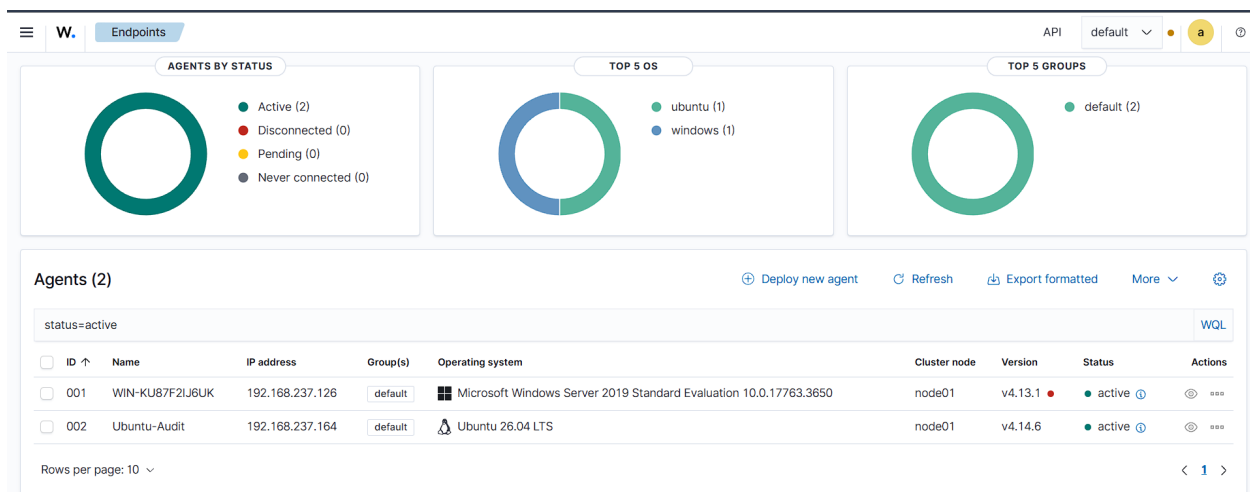
A single-node Wazuh deployment was implemented consisting of the **Wazuh Manager**, **OpenSearch Indexer**, and **Wazuh Dashboard**. The Wazuh Manager was responsible for receiving logs from registered agents, processing security events, and evaluating them against built-in and custom detection rules. Processed alerts were stored in the OpenSearch Indexer and visualized through the Wazuh Dashboard.



6.3 Endpoint Integration

The Windows Server 2019 virtual machine was configured with the Wazuh Agent to forward security events to the Wazuh Manager. To enhance endpoint visibility, Microsoft Sysmon was installed and configured to generate detailed telemetry related to process creation, command-line execution, registry modifications, scheduled tasks, authentication events, and service installations.

Once the agent was successfully registered, the endpoint appeared as an active agent within the Wazuh Dashboard. This confirmed that communication between the endpoint and the SIEM server had been established successfully.



6.4 Log Collection and Event Processing

Security events generated on the Windows endpoint were collected by Sysmon and Windows Event Logs before being forwarded by the Wazuh Agent to the Wazuh Manager. Upon receiving the logs, Wazuh decoded the incoming events, applied correlation rules, and evaluated them against the implemented detection logic.

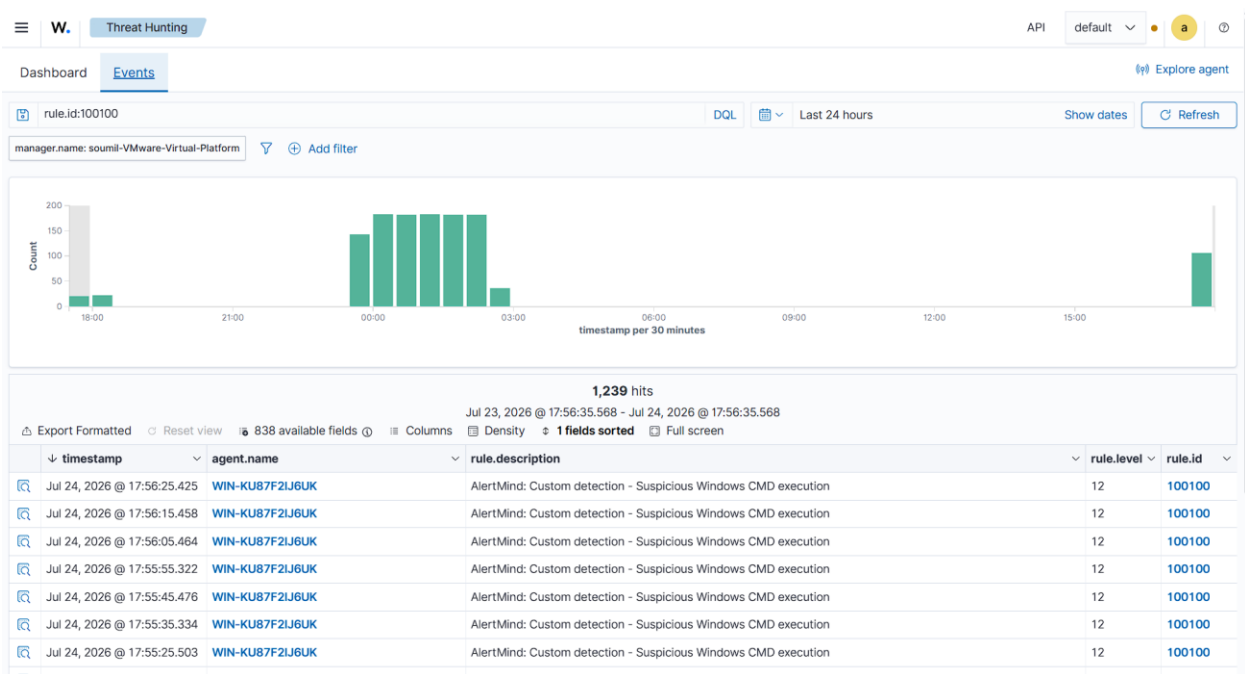
Alerts generated by the SIEM were indexed within OpenSearch and made available through the Wazuh Dashboard. This centralized workflow enabled analysts to review security events, examine affected systems, and identify suspicious activities in real time.

In addition to endpoint monitoring, AWS CloudTrail logs were integrated to demonstrate the collection of cloud activity logs. This expanded the monitoring capability beyond traditional endpoints and illustrated how cloud events can be incorporated into a centralized SIEM environment.

6.5 Dashboard Monitoring

The Wazuh Dashboard served as the primary interface for monitoring alerts and reviewing security events. It provided centralized visibility into endpoint status, generated alerts, rule execution, and MITRE ATT&CK mappings.

The dashboard also enabled filtering based on rule IDs, severity levels, agents, and timestamps, allowing analysts to efficiently investigate suspicious activities. Throughout the project, the dashboard was used to validate custom detection rules and verify the successful execution of Atomic Red Team simulations.



6.6 Cloud Activity Monitoring

To demonstrate monitoring beyond on-premises systems, AWS CloudTrail logs were integrated into the Wazuh platform. CloudTrail records management events performed within an AWS account, such as user authentication, API calls, and configuration changes.

These logs were ingested into Wazuh, allowing cloud events to be monitored alongside endpoint logs through a unified dashboard. Although the project primarily focused on endpoint security monitoring, this integration illustrates how cloud activity can also be incorporated into a centralized SOC environment.

CHAPTER 7

Detection Engineering and Rule Validation

7.1 Overview

Detection engineering is the process of designing, implementing, and validating detection rules to identify malicious activities within an organization's infrastructure. Rather than relying solely on default SIEM rules, custom detections allow organizations to monitor behaviours that are relevant to their security requirements.

For the AlertMind project, a set of **twelve custom detection rules** was developed and integrated into the Wazuh SIEM. These rules were designed to detect common attack techniques associated with command execution, PowerShell misuse, persistence mechanisms, account manipulation, and unauthorized system changes. Each rule was mapped to the corresponding **MITRE ATT&CK** technique to provide standardized classification and improve investigation.

7.2 Detection Strategy

The detection strategy focused on identifying attacker behaviour instead of specific malware signatures. Security events collected from Sysmon and Windows Event Logs were evaluated by the Wazuh Manager using custom XML rules.

The rules were designed to detect suspicious activities commonly observed during the early stages of cyberattacks, including:

- Suspicious Windows Command Shell Activity
- Suspicious Encoded PowerShell Execution
- Suspicious PowerShell Download Cradle Execution
- Multiple failed windows login attempts
- New windows user account creation
- New Windows service installation
- Registry Run Key persistence
- Scheduled task creation
- Successful Remote Logon Detected
- Suspicious Powershell Network Connection
- Startup Folder Execution Detected
- Process Execution from Temp Folder Detected

7.3 Custom Detection Rules

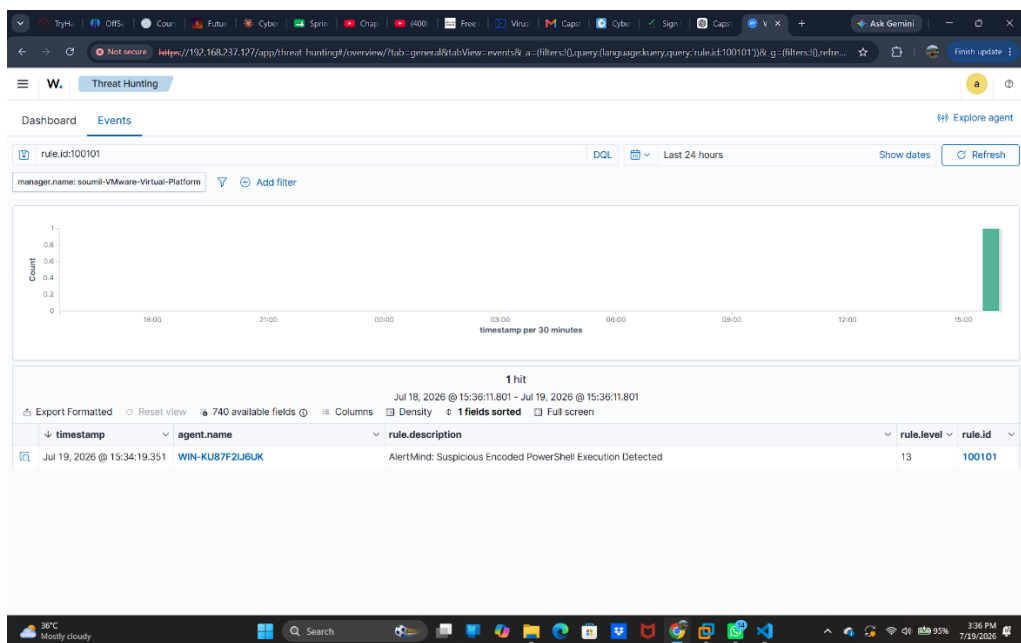
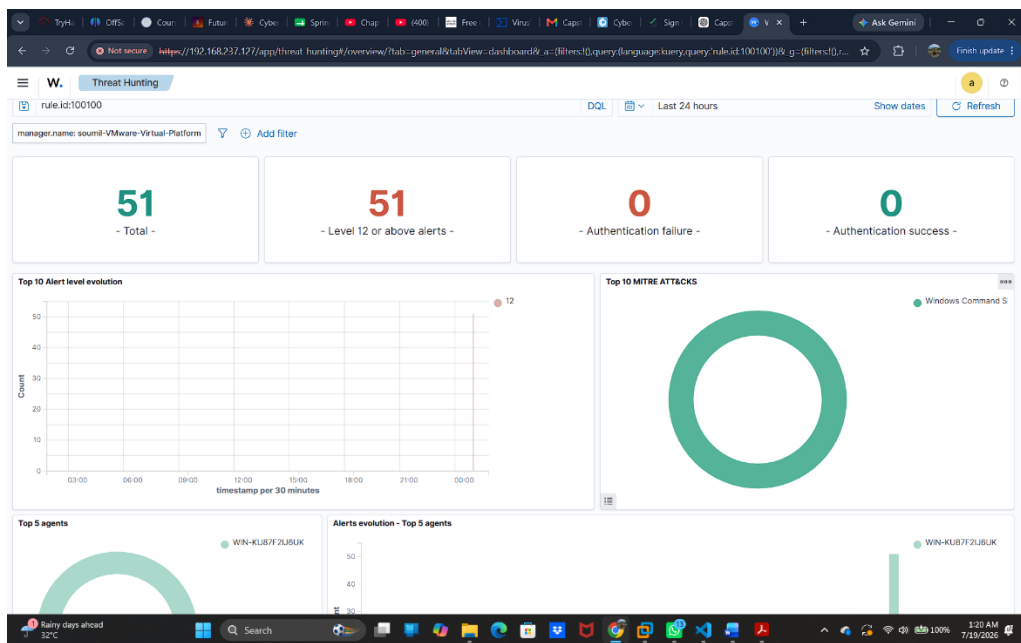
A total of **twelve custom detection rules** were successfully implemented and validated during the project. Each rule targets a specific attack technique and is mapped to the MITRE ATT&CK framework.

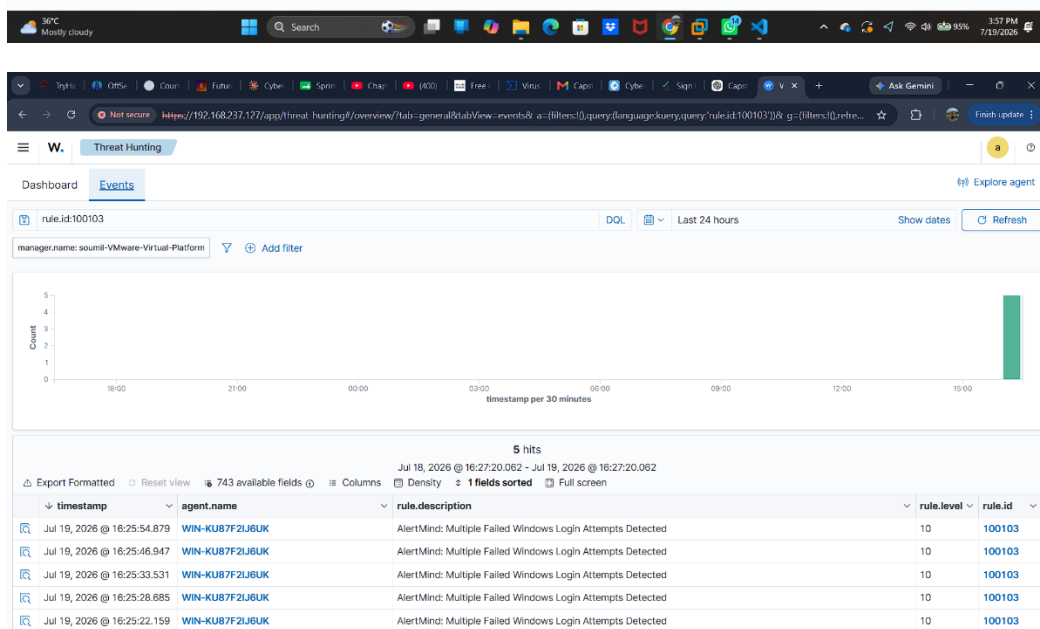
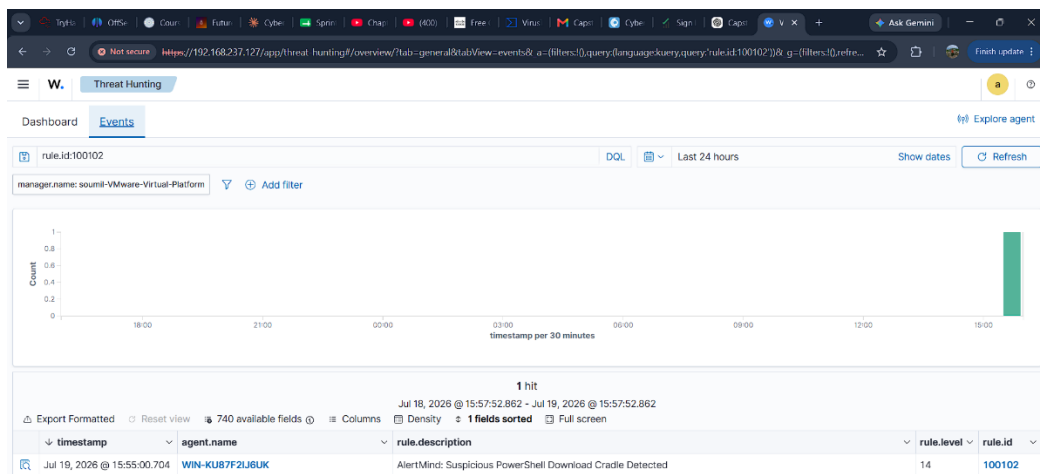
Rule ID	Detection Rule	MITRE ATT&CK Technique
100100	Suspicious Windows Command Shell Activity	T1059.003 – Windows Command Shell

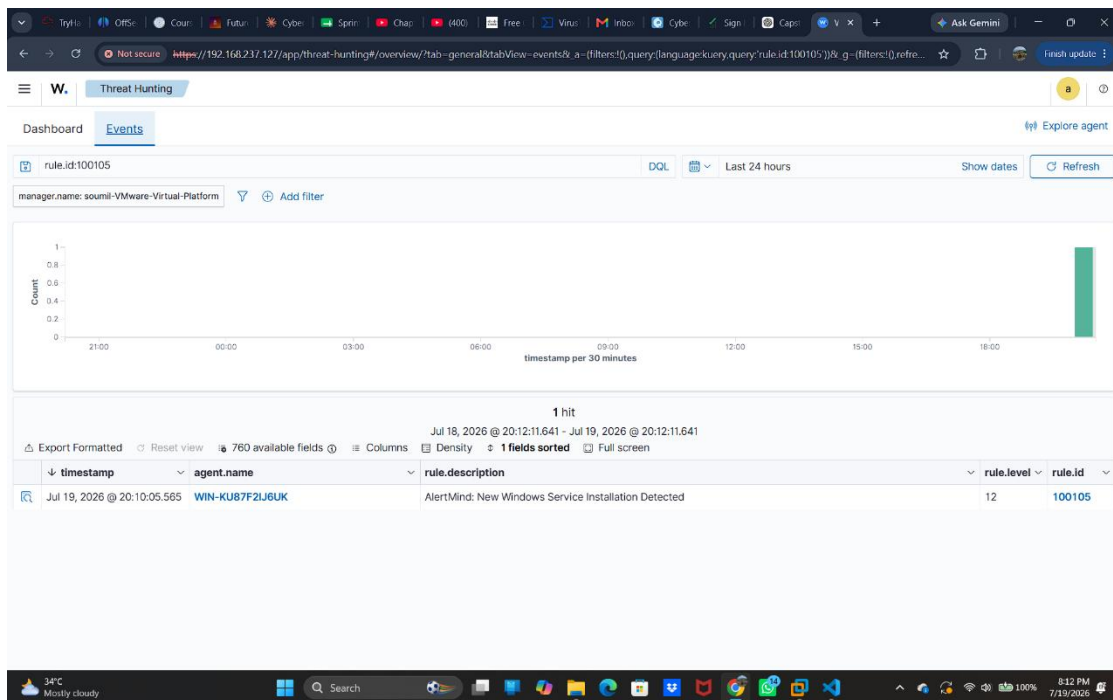
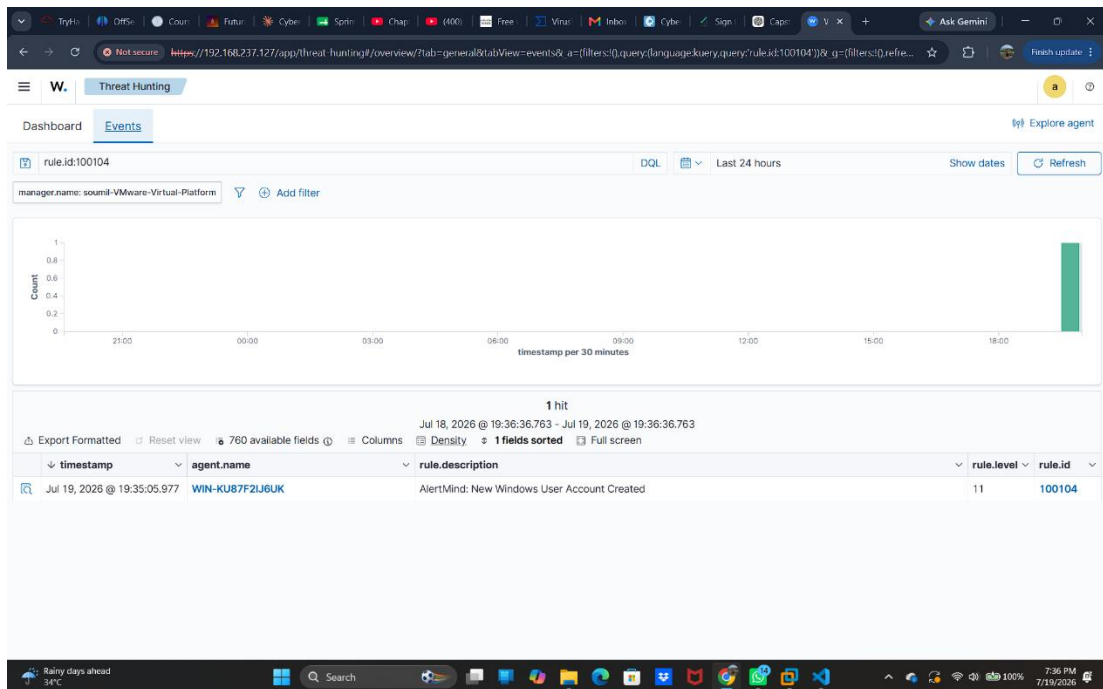
Rule ID	Detection Rule	MITRE ATT&CK Technique
100101	Suspicious Encoded PowerShell Execution	T1059.001 – PowerShell
100102	Suspicious PowerShell Download Cradle Execution	T1059.001 – PowerShell
100103	Multiple Failed Windows Login Attempts	T1110 – Brute Force
100104	New Windows User Account Creation	T1136 – Create Account
100105	New Windows Service Installation	T1543.003 – Windows Service
100106	Registry Run Key Persistence	T1547.001 – Registry Run Keys / Startup Folder
100107	Scheduled Task Creation	T1053.005 – Scheduled Task
100108	Successful Remote Logon Detected	T1021.001 – Remote Services: Remote Desktop Protocol
100109	<i>Rule removed during validation to eliminate redundant detections.</i>	—
100110	Suspicious PowerShell Network Connection	T1105 – Ingress Tool Transfer
100111	Startup Folder Execution Detected	T1547.001 – Registry Run Keys / Startup Folder
100112	Process Execution from Temp Folder Detected	T1204.002 – Malicious File Execution

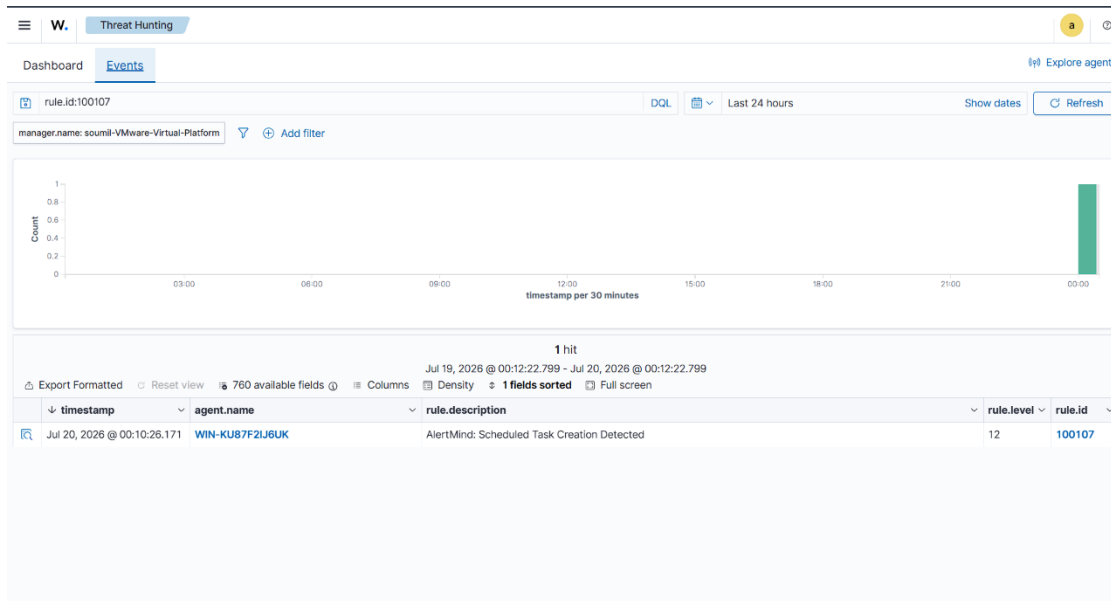
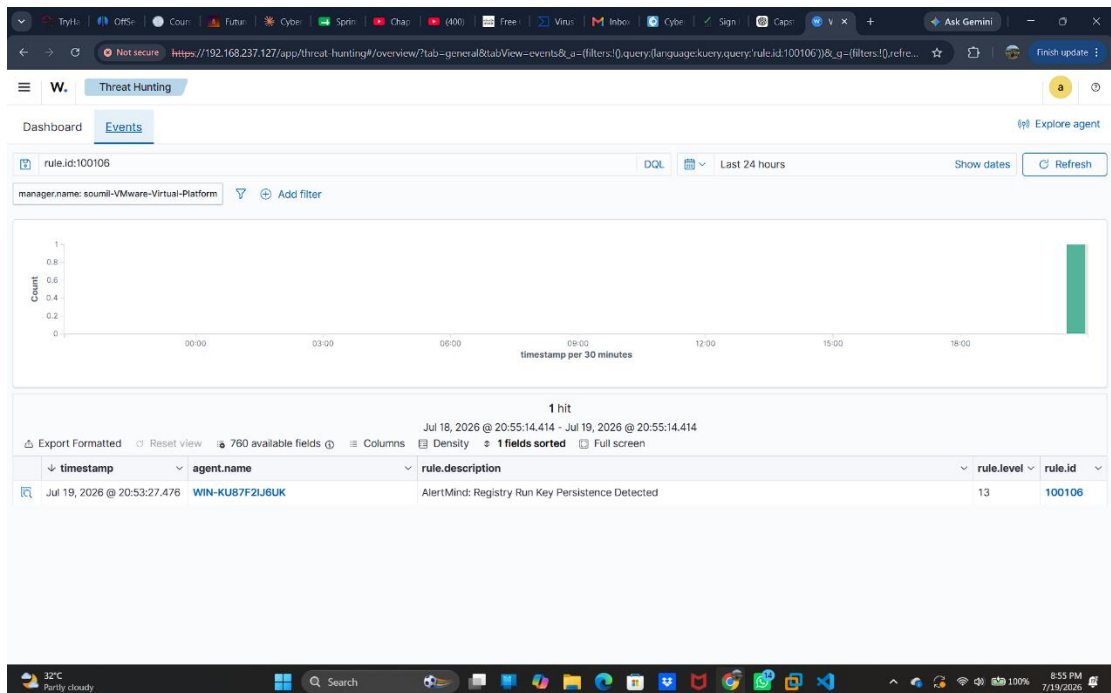
During the development of the custom detection rule set, an initial rule with **Rule ID 100109** was designed to monitor an additional Windows security event. However, during validation, it was observed that the rule either generated redundant alerts or overlapped with the functionality of existing detection rules without providing significant additional security value.

To maintain a concise and effective rule set, Rule **100109** was excluded from the final implementation. The remaining eleven rules provided comprehensive coverage of the targeted attack techniques while minimizing duplicate detections and unnecessary alert generation.









rule.id:100108

DQL

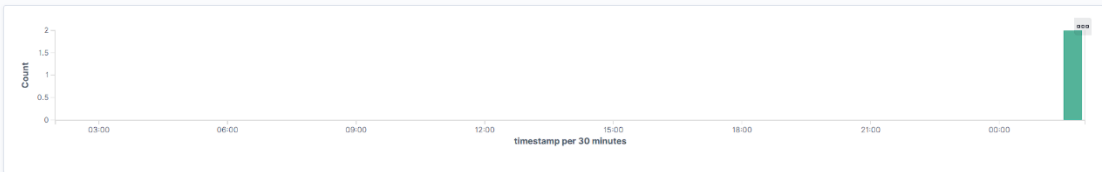
Last 24 hours

Show dates

Refresh

manager.name: soulml-VMware-Virtual-Platform

Add filter



2 hits

Jul 19, 2026 @ 01:59:10.874 - Jul 20, 2026 @ 01:59:10.874

Export Formatted Reset view 766 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 20, 2026 @ 01:58:09.997	WIN-KU87F2J6UK	AlertMind: Successful Remote Logon Detected	11	100108
Jul 20, 2026 @ 01:58:05.870	WIN-KU87F2J6UK	AlertMind: Successful Remote Logon Detected	11	100108

data.win.system.eventID:3

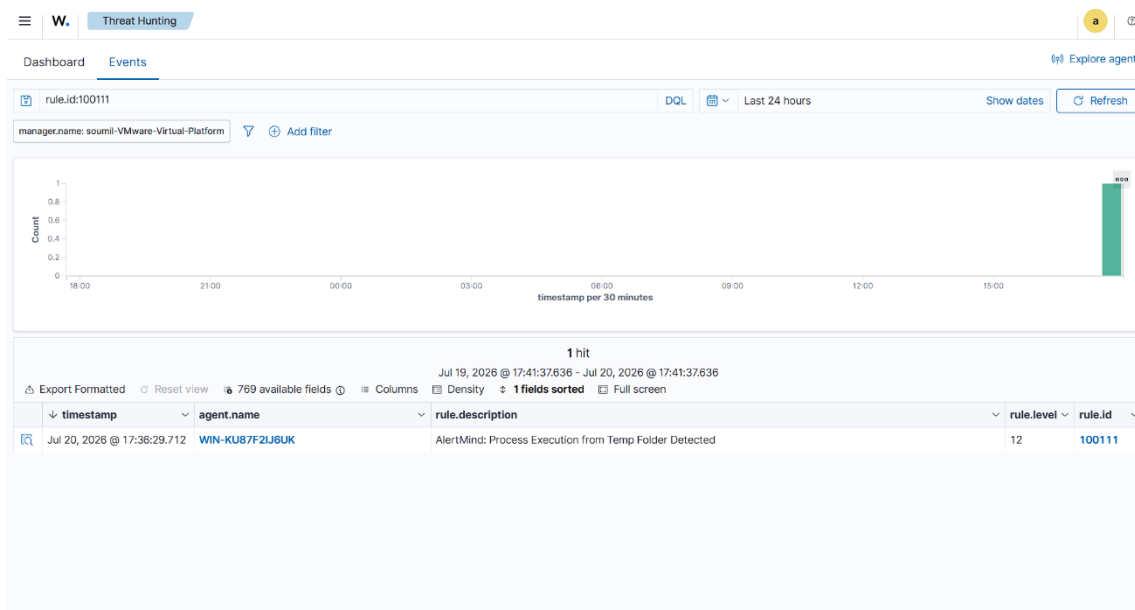
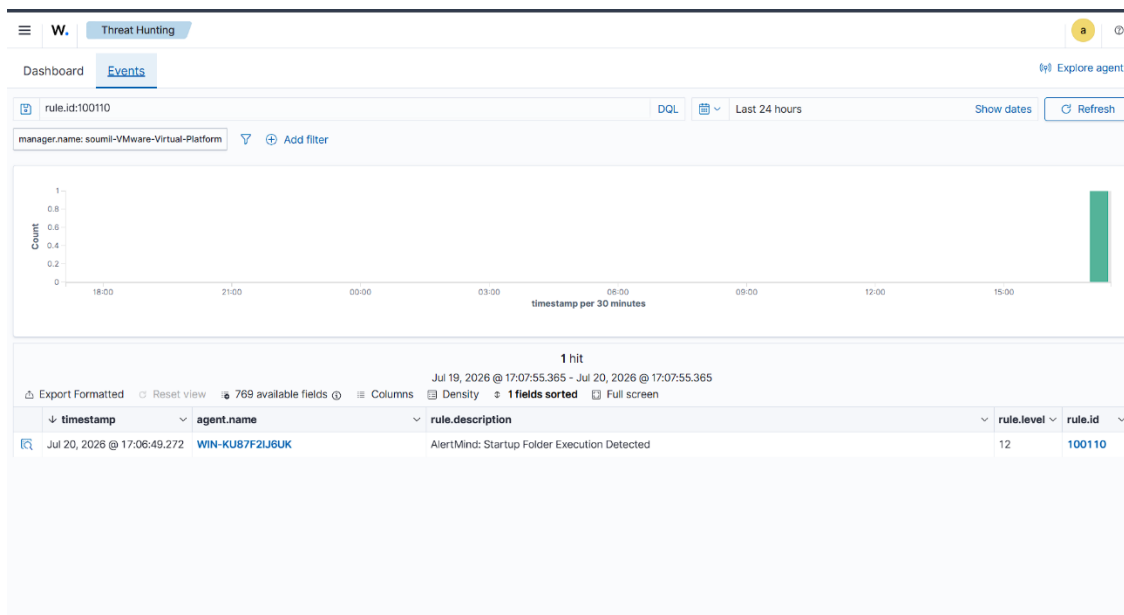
DQL

Refresh

manager.name: soulml-VMware-Virtual-Platform

Add filter

No results match your search criteria

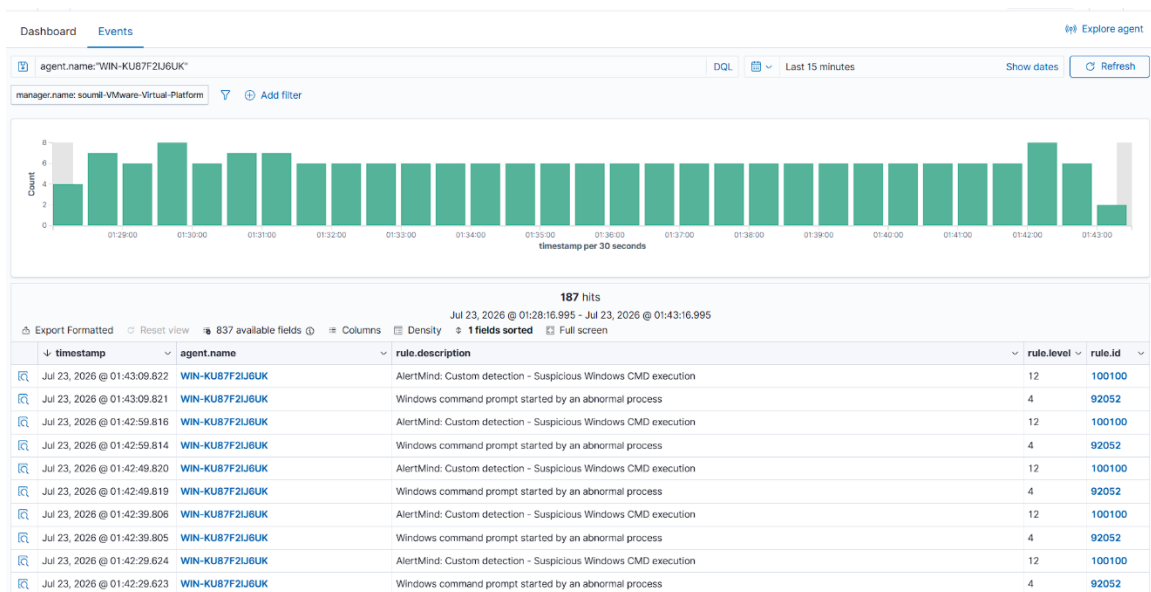


7.4 Rule Validation

To verify the effectiveness of the implemented detections, representative attack scenarios were executed using **Atomic Red Team**. These tests generated realistic security events on the Windows Server endpoint, allowing the corresponding Wazuh rules to trigger alerts.

Successful detection confirmed that the implemented rules were functioning as intended and that endpoint logs were being collected, processed, and correlated correctly by the SIEM.

The generated alerts were reviewed through the Wazuh Dashboard, where the associated rule ID, severity level, affected endpoint, timestamp, and MITRE ATT&CK mapping were displayed.



7.5 Representative Detection Example

One representative example is the detection of suspicious Command Prompt execution. During validation, an Atomic Red Team test simulated command execution on the Windows endpoint. Sysmon recorded the process creation event, which was forwarded to the Wazuh Manager through the Wazuh Agent.

The custom detection rule identified the event as suspicious and generated an alert within the Wazuh Dashboard. The alert was mapped to **MITRE ATT&CK Technique T1059.003 (Windows Command Shell)** and later analyzed by the AlertMind AI Assistant.

This workflow demonstrates how endpoint telemetry, SIEM correlation, and AI-assisted analysis operate together within the proposed architecture.

t id	1784751189.9353388
t input.type	log
t location	EventChannel
t manager.name	soumil-VMware-Virtual-Platform
t rule.description	AlertMind: Custom detection - Suspicious Windows CMD execution
# rule.firedtimes	261
t rule.groups	local, syslog, sshd, alertmind, custom_detection, sysmon
t rule.id	100100
# rule.level	12
rule.mail	true
t rule.mitre.id	T1059.003
t rule.mitre.tactic	Execution
t rule.mitre.technique	Windows Command Shell
timestamp	Jul 23, 2026 @ 01:43:09.822

7.6 Detection Workflow

The implemented detection workflow follows the sequence illustrated below:

1. A security event is generated on the Windows endpoint.
2. Sysmon records the event and forwards it to the Wazuh Agent.
3. The Wazuh Agent sends the event to the Wazuh Manager.
4. Custom detection rules evaluate the incoming log.
5. If suspicious behaviour is identified, Wazuh generates an alert.
6. The alert is displayed in the Wazuh Dashboard and later analyzed by the AlertMind AI Assistant.

This workflow ensures that all alerts undergo standardized detection before being presented to analysts for further investigation.

CHAPTER 8

AI-Assisted Alert Analysis

8.1 Overview

Traditional Security Operations Centers (SOCs) often generate a large number of security alerts, requiring analysts to manually examine each event before determining the appropriate response. This process can be time-consuming, particularly for repetitive or low-priority alerts. To improve the efficiency of alert triage, the AlertMind project incorporates a Large Language Model (LLM) that assists analysts by generating structured summaries and investigation guidance.

The AI component does not perform threat detection or automate incident response. Instead, it operates after the SIEM has generated an alert, providing contextual information to help analysts understand the security event more quickly.

8.2 AlertMind Architecture

The AlertMind AI Assistant was developed as a separate application using **Python**, **LangChain**, and the **Groq API**, which provides access to the **Llama 3.1 8B Instant** language model.

When a security alert is generated by Wazuh, the alert data is exported in JSON format and supplied to the AI assistant. The application extracts relevant information from the alert, constructs a structured prompt, and submits it to the language model for analysis. The generated response is then presented to the analyst for review.

This modular design ensures that the AI assistant operates independently of the SIEM infrastructure, allowing future upgrades or model replacements without affecting the detection engine.

8.4 AI-Generated Output

For each processed alert, the AlertMind assistant generates a structured response containing:

- A concise summary of the detected activity.
- The associated MITRE ATT&CK technique.
- Possible security implications.
- Suggested investigation steps.
- Recommended response actions.
- A draft notification that can be reviewed before communication with stakeholders.

This standardized output reduces the effort required to interpret raw SIEM alerts and assists analysts in prioritizing their investigations.

```
=====
2026-07-24 02:14:38.405928
Alert File: alerts/powershell.json

Rule ID: 100100
Description: AlertMind: Custom detection - Suspicious Windows CMD execution
Agent: WIN-KU87F2I36UK
Timestamp: 2026-07-23T17:21:50.814+0530
MITRE ID: T1059.003
MITRE Technique: Windows Command Shell
MITRE Tactic: Execution
Command: "C:\Program Files\Splunk\bin\splunk.exe" show-decrypted --value '$7$lu+L9s+SVI0l/g1gmALL80BG73litisQLc8oCdq1pcyrpLq0vD6Q=='
Parent Process: C:\Windows\System32\cmd.exe
User: NT SERVICE\Splunkd

(venv) soumil@soumil-Virtual-Platform:~/AlertMind-LLMs
```

```
=====
2026-07-24 02:14:39.353223
**Alert Summary**

A suspicious Windows CMD execution was detected on WIN-KU87F2I36UK at 2026-07-23T17:21:50.814+0530. The command executed was related to Splunk, and it was run by the NT SERVICE\Splunkd user. The parent process was cmd.exe.

**MITRE ATT&CK Technique and Tactic**

- MITRE ID: T1059.003
- MITRE Technique: Windows Command Shell
- MITRE Tactic: Execution

**Suggested Investigation Queries**

1. What is the purpose of the Splunk command being executed?
2. Is the NT SERVICE\Splunkd user a legitimate service account?
3. Is the cmd.exe parent process legitimate, or is it a potential indicator of compromise?

**Recommended Analyst Actions**

1. Review the Splunk configuration and logs to understand the purpose of the command.
2. Verify the legitimacy of the NT SERVICE\Splunkd user and its permissions.
3. Investigate the cmd.exe parent process to determine its legitimacy.

**Draft Notification to the affected user**

Subject: Security Alert: Suspicious Windows CMD Execution

Dear Splunk Administrator,

A security alert has been triggered on WIN-KU87F2I36UK due to a suspicious Windows CMD execution related to Splunk. The command was executed by the NT SERVICE\Splunkd user at 2026-07-23T17:21:50.814+0530.

We are investigating this incident to determine its legitimacy. In the meantime, please review the Splunk configuration and logs to understand the purpose of the command.

If you have any questions or concerns, please do not hesitate to contact the security team.

Best regards,
AlertMind Security Team
(venv) soumil@soumil-Virtual-Platform:~/AlertMind-LLMs
```

8.5 Responsible AI Considerations

To ensure responsible use of artificial intelligence within the SOC environment, several safeguards were incorporated into the AlertMind assistant.

Before being submitted to the language model, user inputs are sanitized to reduce the risk of prompt injection and malformed requests. Both prompts and AI responses are logged to support auditing and troubleshooting. Additionally, each generated response includes a disclaimer indicating that the recommendations are intended to assist analysts and should not replace human judgement.

No automated actions are performed based on AI output. Final decisions regarding investigation and incident response remain the responsibility of the SOC analyst.

CHAPTER 9

Dashboard Visualization and Monitoring

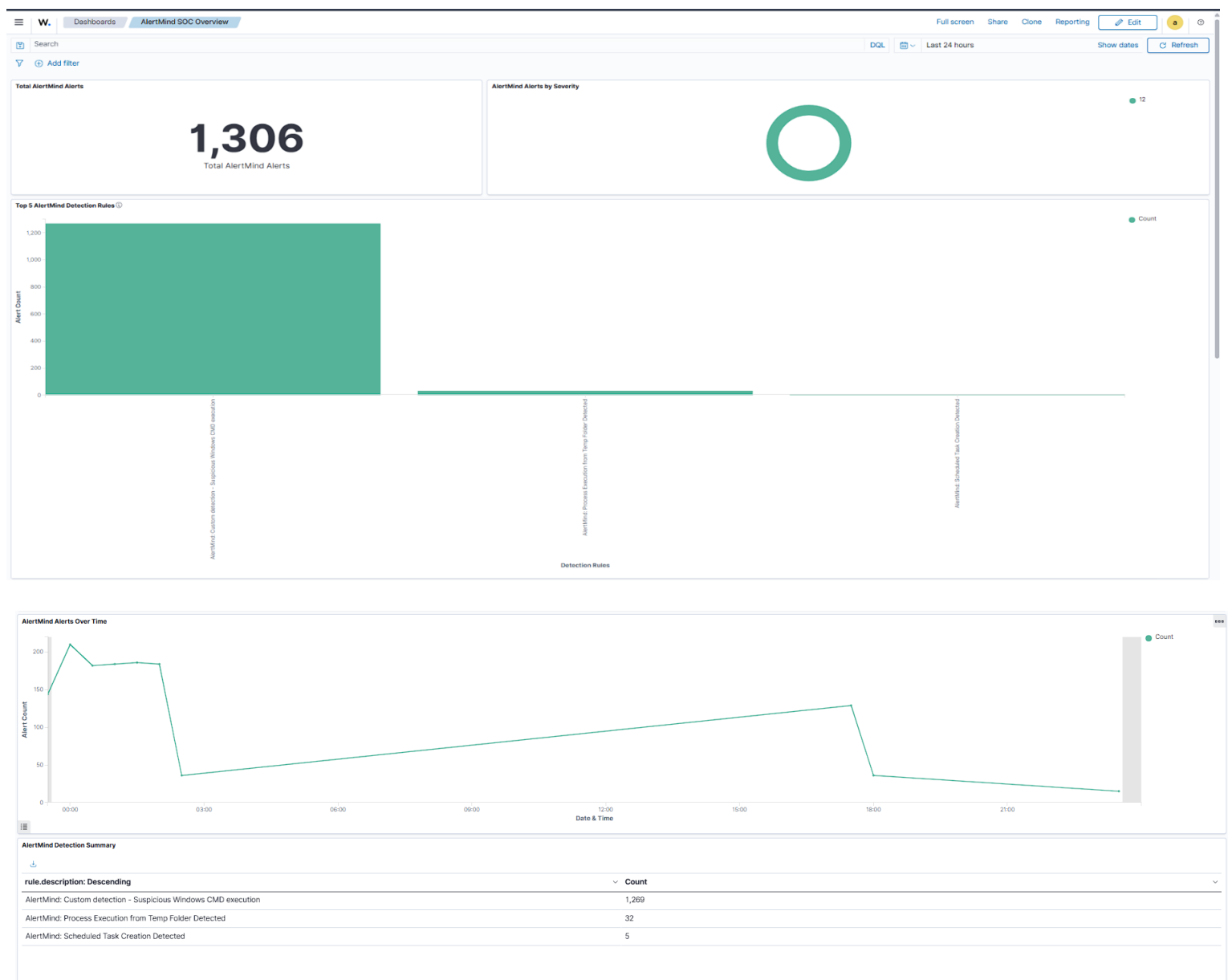
9.1 Overview

Effective visualization is an essential component of a Security Operations Centre (SOC), enabling analysts to monitor security events and identify suspicious activities efficiently. In the AlertMind project, the Wazuh Dashboard served as the primary interface for viewing alerts, monitoring endpoint status, and analyzing detection results. Custom dashboards were created to provide a centralized view of security events generated by the implemented detection rules.

9.2 SOC Overview Dashboard

A custom **SOC Overview Dashboard** was developed to display key security metrics within the monitoring environment. The dashboard presents information such as the total number of alerts, alert severity distribution, active monitored agents, and recent security events.

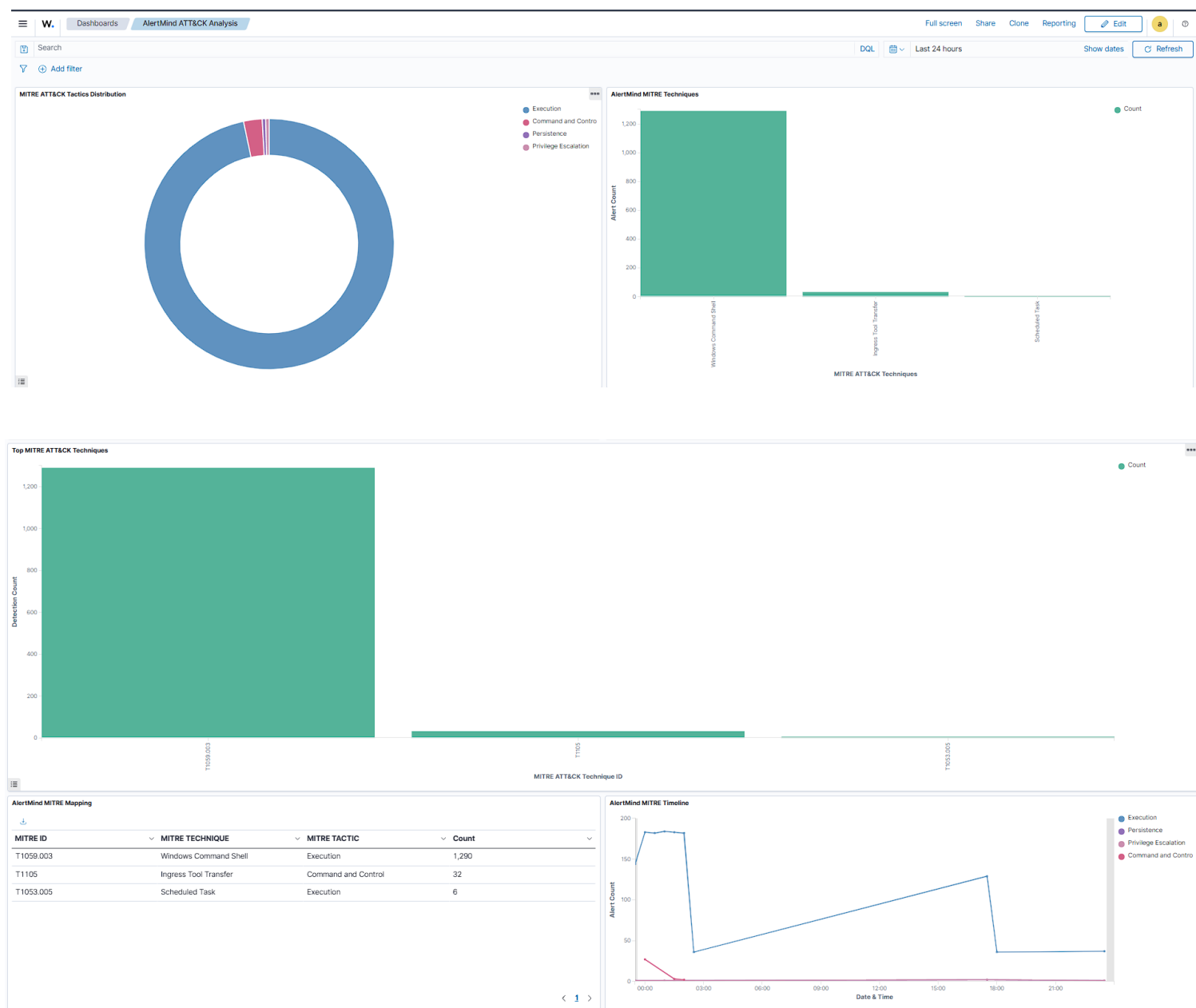
By consolidating these metrics into a single interface, analysts can quickly assess the overall security status of the environment and identify areas requiring further investigation.



9.3 MITRE ATT&CK Dashboard

To improve threat analysis, a dedicated dashboard was created to visualize alerts according to the **MITRE ATT&CK framework**. Each triggered detection rule was mapped to its corresponding ATT&CK technique, allowing analysts to understand the tactics and techniques associated with observed activities.

This visualization provides additional context for security events and supports more structured incident investigation.



9.4 Monitoring Benefits

The dashboards developed for AlertMind provide several operational advantages:

- Centralized monitoring of security events.
- Real-time visualization of alerts and endpoint activity.
- Improved understanding through MITRE ATT&CK mapping.
- Faster identification of suspicious behaviour.

- Simplified investigation using graphical summaries.

These dashboards complement the AI assistant by providing analysts with both visual insights and AI-generated contextual analysis.

CHAPTER 10

Testing and Validation

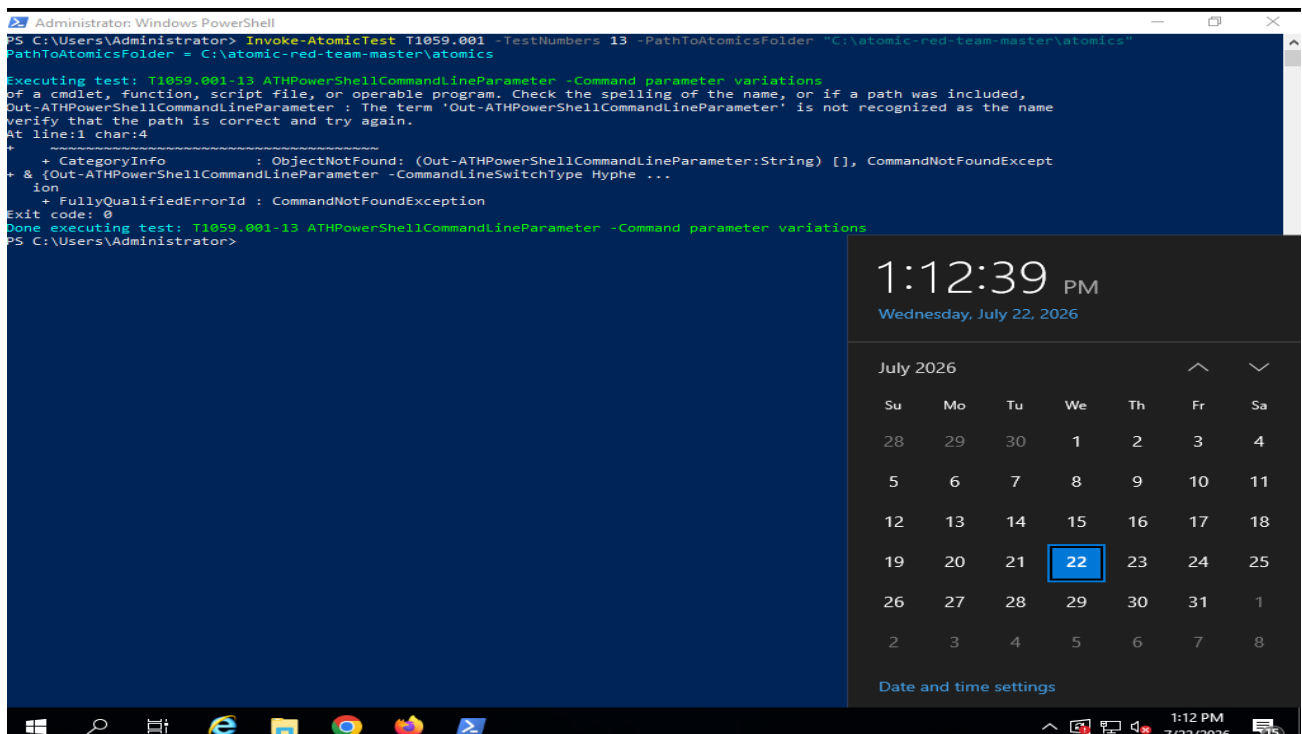
10.1 Overview

To verify the effectiveness of the proposed AlertMind Security Operations Centre, a series of controlled security tests were conducted within the virtual laboratory environment. The objective was to validate that the Wazuh SIEM successfully detected simulated attack activities and that the AlertMind AI Assistant correctly generated contextual analysis for the detected alerts.

10.2 Testing Methodology

The testing methodology consisted of four sequential stages:

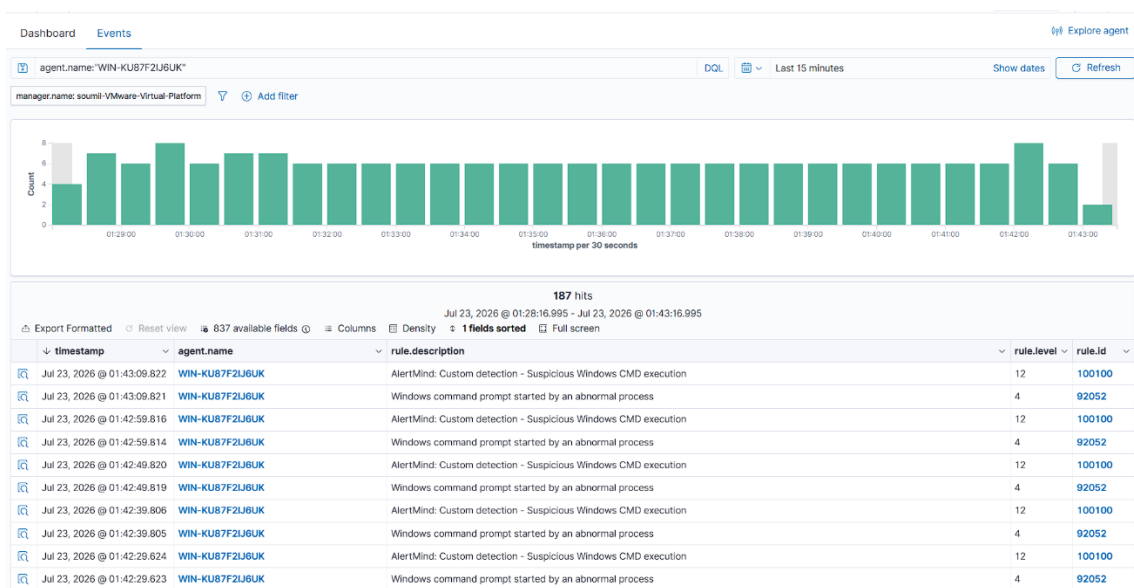
1. Simulating attacker behaviour using Atomic Red Team tests.
2. Collecting endpoint telemetry through Sysmon and the Wazuh Agent.
3. Processing security events using the custom Wazuh detection rules.
4. Analyzing the generated alerts using the AlertMind AI Assistant.



10.3 Detection Validation

Each implemented detection rule was tested by executing its corresponding attack simulation or security event on the Windows Server endpoint. The generated events were collected by the Wazuh Agent, processed by the Wazuh Manager, and displayed in the Wazuh Dashboard.

The successful generation of alerts confirmed that the custom detection rules correctly identified the targeted security activities.



```
=====
2026-07-24 02:14:38.405928
Alert File: alerts/powershell.json

Rule ID: 100100
Description: AlertMind: Custom detection - Suspicious Windows CMD execution
Agent: WIN-KU87F2U6UK
Timestamp: 2026-07-23T17:21:50.814+0530
MITRE ID: T1059.003
MITRE Technique: Windows Command Shell
MITRE Tactic: Execution
Command: "C:\\Program Files\\Splunk\\bin\\splunk.exe" show-decrypted --value \"$7$luN+L9s+SVIQ0l/g1gmALL00BG73l1tisQLc0oGdq1pcyrLq0vD6Q==\"
Parent Process: C:\\Windows\\System32\\cmd.exe
User: NT SERVICE\\Splunkd

(venv) soumil@soumil-VMware-Virtual-Platform:~/AlertMind-LLMS
```

10.4 AI Assistant Validation

After each alert was generated, the AlertMind AI Assistant processed the exported Wazuh JSON alert and produced structured security analysis.

For every tested alert, the assistant successfully generated:

- Alert summary
- MITRE ATT&CK mapping

- Investigation recommendations
- Response guidance
- Draft notification

The AI assistant consistently produced structured outputs without modifying the original detection results generated by Wazuh.

```
=====
2026-07-24 02:14:39.353223
**Alert Summary**

A suspicious Windows CMD execution was detected on WIN-KU87F2I36UK at 2026-07-23T17:21:50.814+0530. The command executed was related to Splunk, and it was run by the NT SERVICE\Splunkd user. The parent process was cmd.exe.

**MITRE ATT&CK Technique and Tactic**
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- MITRE Technique: Windows Command Shell
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**Suggested Investigation Queries**
1. What is the purpose of the Splunk command being executed?
2. Is the NT SERVICE\Splunkd user a legitimate service account?
3. Is the cmd.exe parent process legitimate, or is it a potential indicator of compromise?

**Recommended Analyst Actions**
1. Review the Splunk configuration and logs to understand the purpose of the command.
2. Verify the legitimacy of the NT SERVICE\Splunkd user and its permissions.
3. Investigate the cmd.exe parent process to determine its legitimacy.

**Draft Notification to the affected user**

Subject: Security Alert: Suspicious Windows CMD Execution

Dear Splunk Administrator,

A security alert has been triggered on WIN-KU87F2I36UK due to a suspicious Windows CMD execution related to Splunk. The command was executed by the NT SERVICE\Splunkd user at 2026-07-23T17:21:50.814+0530.

We are investigating this incident to determine its legitimacy. In the meantime, please review the Splunk configuration and logs to understand the purpose of the command.

If you have any questions or concerns, please do not hesitate to contact the security team.

Best regards,
AlertMind Security Team
(venv) soumil@soumil-Virtual-Platform:~/AlertMind-LLMs
```

CHAPTER 11

Results and Discussion

11.1 Overview

The implementation and testing of the AlertMind AI-Assisted Security Operations Centre demonstrated that the proposed architecture successfully integrates centralized security monitoring with AI-assisted alert analysis. Throughout the evaluation, security events generated on the Windows Server endpoint were collected by the Wazuh Agent, processed by the Wazuh Manager, and displayed through the Wazuh Dashboard. The generated alerts were then analyzed by the AlertMind AI Assistant, providing additional context to support analyst decision-making.

11.2 Results

The validation process confirmed that the implemented custom detection rules successfully identified the simulated attack activities executed during testing. Alerts were generated for

techniques including command execution, PowerShell misuse, failed login attempts, persistence mechanisms, service installation, scheduled task creation, and other suspicious system activities.

The dashboard provided centralized visibility into detected events, while the AlertMind AI Assistant produced structured outputs containing alert summaries, MITRE ATT&CK mappings, investigation guidance, and recommended response actions. This demonstrated that the AI component could effectively assist analysts without altering the underlying detection process performed by the SIEM.

11.3 Discussion

The project demonstrates the benefits of combining a traditional SIEM with a Large Language Model to improve security operations. Wazuh remained responsible for reliable event detection through behaviour-based rules, while the AI assistant reduced the effort required to interpret alerts by presenting security information in a structured and consistent format.

Although the prototype successfully achieved its objectives, several limitations remain. The system was evaluated within a controlled virtual laboratory and therefore does not reflect the scale or complexity of a production enterprise environment. Additionally, the AI assistant provides recommendations only and does not perform automated response actions, ensuring that all security decisions remain under human supervision.

Despite these limitations, the results indicate that AI-assisted alert analysis can improve the efficiency of SOC workflows by supporting faster understanding of security events while maintaining analyst oversight.

CHAPTER 12

Conclusion

This project presented **AlertMind**, an AI-Assisted Mini Security Operations Centre (SOC) that combines the capabilities of the Wazuh Security Information and Event Management (SIEM) platform with a Large Language Model to support security analysts during alert investigation.

The implemented system successfully demonstrated centralized log collection, behaviour-based threat detection, dashboard visualization, and AI-assisted alert analysis within a virtual laboratory environment. Custom detection rules were developed to identify a range of common attack techniques, and their effectiveness was validated through controlled attack simulations. The generated alerts were further analyzed by the AlertMind AI Assistant, which produced structured summaries, MITRE ATT&CK mappings, investigation guidance, and response recommendations.

A key contribution of this project is the integration of AI as a decision-support tool rather than a replacement for traditional detection mechanisms. Wazuh remained responsible for threat detection, while the AI assistant enhanced analyst efficiency by simplifying the interpretation of

security alerts. The results demonstrate that combining rule-based detection with AI-assisted analysis can improve SOC workflows while maintaining human oversight over security decisions.

Overall, the project achieved its objectives and provides a practical proof of concept for applying artificial intelligence to support modern Security Operations Centres.

References

- ⇒ MITRE ATT&CK
- ⇒ NIST Cybersecurity Framework 2.0
- ⇒ Wazuh Documentation
- ⇒ Microsoft Sysmon Documentation
- ⇒ Atomic Red Team
- ⇒ LangChain Documentation
- ⇒ Groq API Documentation
- ⇒ Meta Llama 3.1 Documentation
- ⇒ OWASP
- ⇒ CIS Benchmarks
- ⇒ CERT-In Advisories

